Professional Summary

Senior AI Engineer with **7+ years of experience** designing and deploying production Generative AI systems. Core expertise in **RAG architectures**, **Agentic AI**, **Generative UI**, and enterprise-grade LLM integration. Strong Full-Stack delivery capability (Python, React/Next.js) enabling end-to-end AI product development — from model orchestration to user interface. Deployed at scale for a Tier-1 European FinTech (500K+ users), industrial IoT manufacturers, and institutional/public-sector organizations. **66+ technical certifications** including AWS, Microsoft Azure, Google, Oracle, IBM, MongoDB, and Neo4j. M.Sc. in Management Engineering (Digital Transformation) with research on Generative UI and embedding-free RAG systems.

Professional Experience

Independent AI Consultancy — 15+ Projects Delivered **Rome, Italy**
Senior AI Engineer & Technical Consultant *Jan 2020 – Present*
Technical lead for AI architecture design and full-stack implementation across engagements for mid-to-large enterprises. Led cross-functional teams of 2–5 engineers. Client portfolio includes a **Tier-1 European FinTech** (500K+ users), **industrial IoT manufacturers**, and **Italian public-sector cultural institutions**.

- **Institutional AI Deployment (Current):** Supporting a public-sector cultural pilot involving six museums in the Italian institutional museum system. Leading technical work across local-inference AI strategy, non-closed model approaches, backend/API architecture, database design, and cloud infrastructure planning. Contributing to a controlled, privacy-aware deployment model designed for institution-oriented adoption.
- **Agentic AI & Generative UI:** Architected a multi-agent OmniApp with autonomous AI agents and Generative UI, rendering React components in real-time via LLM orchestration (LangChain, Vercel AI SDK). Implemented fallback handling, streaming responses, and tool-use patterns. **Reduced UI development time by 40%.**
- **Enterprise RAG Systems:** Designed and deployed hybrid RAG pipelines (vector search + BM25 + iterative disambiguation) processing 50K+ documents for enterprise knowledge bases. **Reduced hallucination rate by 30%** on customer-facing chatbots.
- **Full-Stack Platform Delivery:** Built 5+ enterprise dashboards (React, Redux, Context API) with high-performance backends (Python FastAPI, Django). Achieved **P95 latency under 800ms** through Redis caching and CDN optimization.
- **Legacy Modernization:** Led COBOL-to-Python migration of approximately 50K lines of code using AI-assisted translation workflows. **Improved rewrite velocity by 60%.**
- **AI-Augmented Development:** Integrated AI coding assistants into daily workflows for development, debugging, code review, and automated test generation, significantly accelerating delivery cycles.
- **Cloud & DevOps:** Designed CI/CD pipelines (GitHub Actions), containerized deployments (Docker, Kubernetes), and cloud infrastructure on Azure and AWS. Applied OWASP security standards.

Freelance **Rome, Italy**
Full-Stack Developer *2018 – 2019*
Started career building web applications and delivering end-to-end solutions for small business clients.

- Built 8+ client projects: responsive web applications (JavaScript, HTML/CSS, PHP), REST APIs (Node.js), and database-driven backends (MySQL, PostgreSQL).
- Managed full project lifecycle from requirements gathering through deployment and maintenance.
- Developed strong foundations in software architecture and client communication that enabled transition to AI-focused engineering.

Key AI Projects

OmniApp — Generative UI Platform: Multi-agent system rendering dynamic React interfaces in real-time via LLM orchestration. Stack: Next.js, Vercel AI SDK, LangChain. **40% faster time-to-feature** vs. traditional development.

Enterprise RAG Engine: Semantic search and Q&A system for 50K+ enterprise documents. Hybrid vector + BM25 retrieval with iterative disambiguation. Stack: Pinecone, ChromaDB, Python. **30% improvement in answer accuracy.**

Industrial IoT Dashboard: Real-time monitoring platform for industrial sensor data. Stack: React, FastAPI, WebSocket, TimescaleDB. Sustained **10K+ data points/sec** with sub-200ms latency.

Technical Skills

AI & GenAI: LLM Integration (OpenAI, Anthropic, Cohere), LangChain, LangGraph, Vercel AI SDK, RAG (Vector + BM25 Hybrid), Agentic AI, CrewAI, Generative UI, Prompt Engineering, LLMOps, AI Evaluation & Guardrails

ML & Data: Python, Pandas, NumPy, Scikit-learn, Keras, Pinecone, ChromaDB, Neo4j

Full-Stack: React / Next.js, TypeScript, Angular, Vue.js, Tailwind CSS, Django, FastAPI, Node.js, Express, GraphQL, REST

Cloud & DevOps: AWS, Microsoft Azure, Docker, Kubernetes, GitHub Actions, Terraform, CI/CD

Databases: PostgreSQL, MongoDB, Pinecone, ChromaDB, Neo4j, TimescaleDB, Redis

Education

Università Unimercatorum

M.Sc. in Management Engineering — Digital Transformation, GPA: 27/30 — ***Full-time available (thesis defense***

- **Thesis:** Framework for Generative UI and embedding-free RAG — reducing computational cost and latency in enterprise AI systems.
- **Research:** 2 papers submitted to international peer-reviewed journals.

Università Niccolò Cusano — **Italy**

B.Sc. in Management Engineering (L-9) — *2018 – 2021*

Thesis on applied machine learning: K-Nearest Neighbors (KNN) algorithm for classification tasks.

Certifications (Selection from 66+)

AI & GenAI: AWS Certified Generative AI Developer – Professional · Microsoft Azure AI Engineer · Oracle OCI Generative AI Professional · IBM Multimodal GenAI Apps · Oracle AI Agent Studio

Data & ML: IBM Data Science Professional · Oracle OCI Data Science Professional · Neo4j Graph Data Science · Google Data Analytics

Database: MongoDB Associate Developer · Oracle AI Vector Search Professional · Neo4j Certified Professional

Security: Google Cybersecurity Professional · Microsoft Security Foundations · Oracle OCI Foundations

Complete list of 66+ certifications available at: `github.com/giulio-leone/cv`

Languages

Italian: Native speaker

English: C1 — Advanced — *Daily professional use in international projects*

EU citizen — Open to UAE work visa sponsorship — Immediately available for relocation (0 days notice).